

Why Latent Chain-of-Thought Helps Exploration but Hurts Precise Computation

Jiaxuan Zou^{1*} · Yaozhong Xiong^{2**} · Yong Liu³ (*equal contribution)

¹ Department of Mathematics and Statistics, Xi'an Jiaotong University

² School of Mathematics, Renmin University of China · ³ Department of Artificial Intelligence, Gaoling School of Artificial Intelligence, Renmin University of China



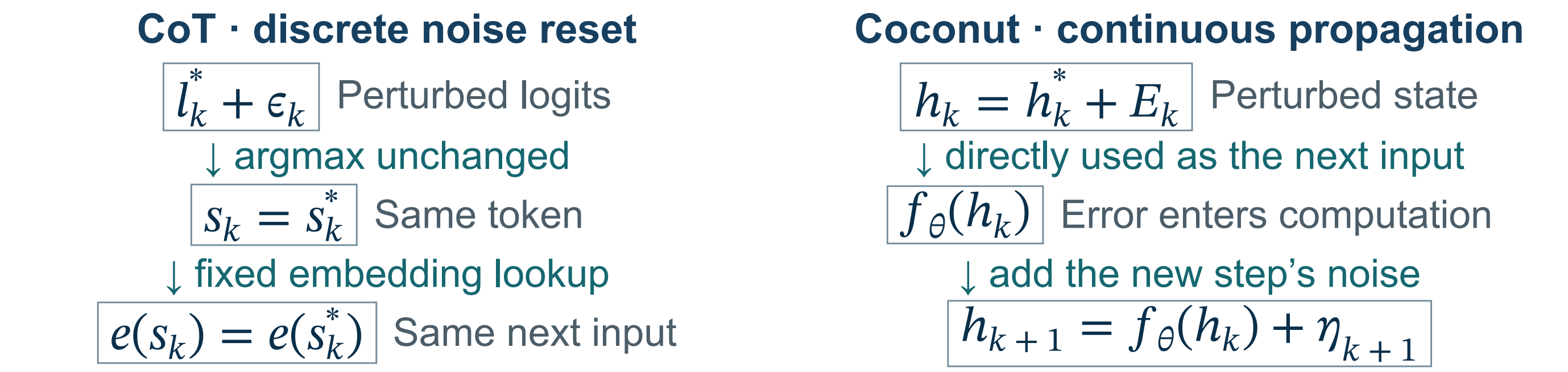
01 Motivation · opposite task effects

Why does Coconut help exploration yet hurt precise computation?

Compared with CoT: −8.8 percentage points on GSM8K; +19.5 points on ProsQA.

02 CoT vs. Coconut

CoT feeds fixed token embeddings; Coconut feeds continuous hidden states.



03 Valid branches & exploration divergence

In a reasoning DAG, $N_{\text{valid}}(v)$ contains B valid successors. The ideal prior is uniform: $q_{\text{PR}}(u \mid v) = 1 / B$.

$$D_{\text{KL}}(q_{\text{PR}} \parallel p) = -\log B - \frac{1}{B} \sum_{u \in N_{\text{valid}}(v)} \log p(u \mid v)$$

If one valid branch has $p(u \mid v) \rightarrow 0$, then $D_{\text{KL}} \rightarrow \infty$: **concentration can suppress exploration.**

04 Conditional Information Bottleneck

Compress the past; preserve information needed to predict the future. Latent prefixes replace explicit reasoning while the remaining suffix supplies supervision.

Theorem 4.1 · Coconut-CIB Duality

Under the constraint of any finite model capacity, the optimization objective of the Coconut curriculum (Preliminaries 3) at stage k can be rigorously reformulated as a constrained optimization problem. Its Lagrangian dual is precisely the Conditional Information Bottleneck (CIB) problem. Specifically:

Primal Problem: The optimization process of Coconut is equivalent to solving:

$$\begin{aligned} \min_{p(h_k \mid X)} \quad & H(S^{(k+1 \dots M)} \mid h_k, X) \\ \text{s. t.} \quad & I(h_k; S^{(1 \dots k)} \mid X) \leq R \end{aligned}$$

Dual:

$$\min_{p(h_k \mid X)} \left[I(h_k; S^{(1 \dots k)} \mid X) - \beta(k) I(h_k; S^{(k+1 \dots M)} \mid X) \right]$$

where $\beta(k) > 0$ ideally satisfies $\beta(k) \sim \frac{k}{M-k}$.
Realizable decoder, finite suffix entropy, positive multiplier; finite β for $k < M$.

10 Empirical evidence · concentration and noise tolerance

A · Candidate-probability trajectory · GSM8K

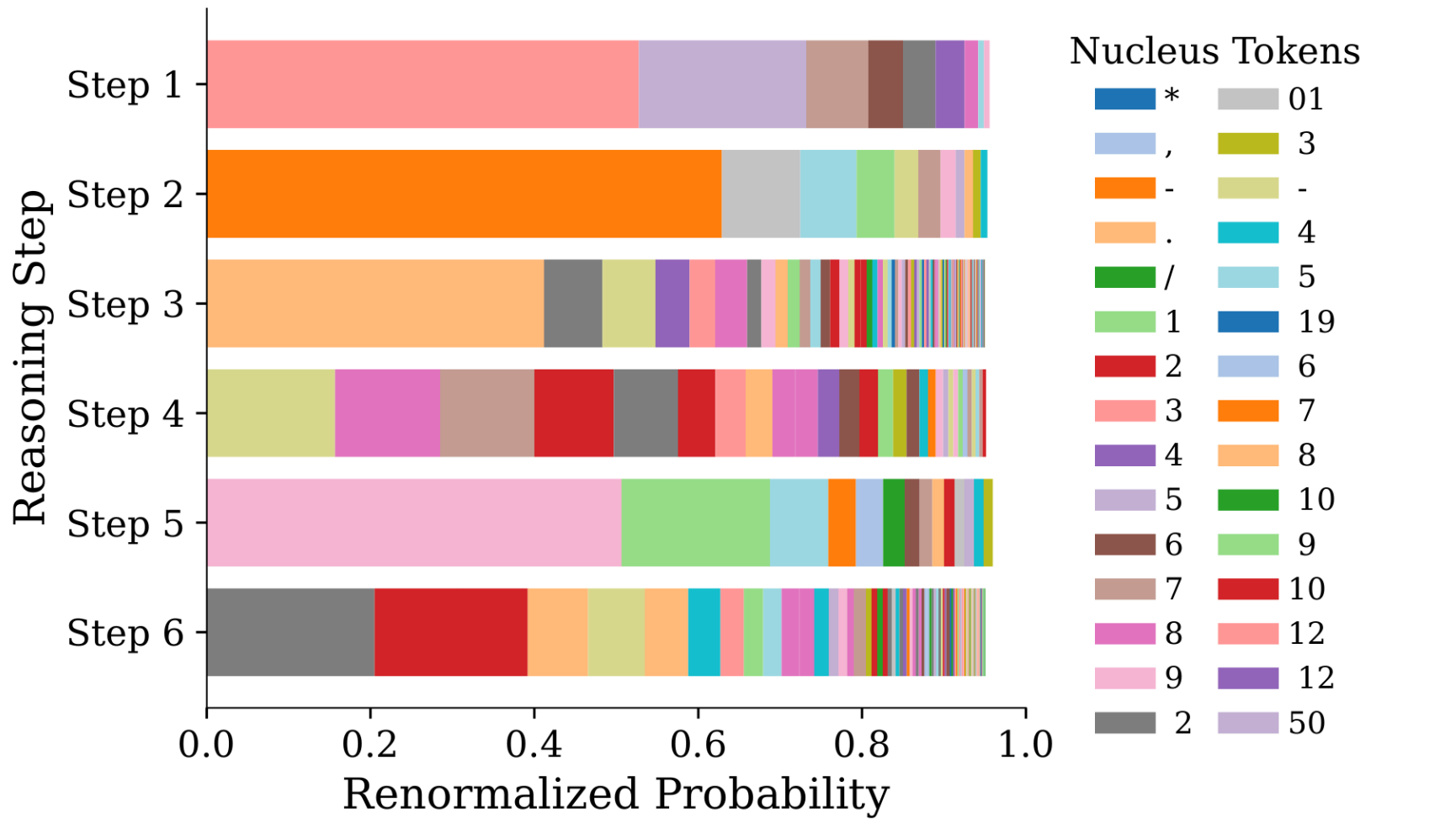


Figure 1 · Six latent reasoning steps on GSM8K.
Multiple candidates retain probability mass; the leading tokens change across steps.

C · Noise response reveals the reset mechanism

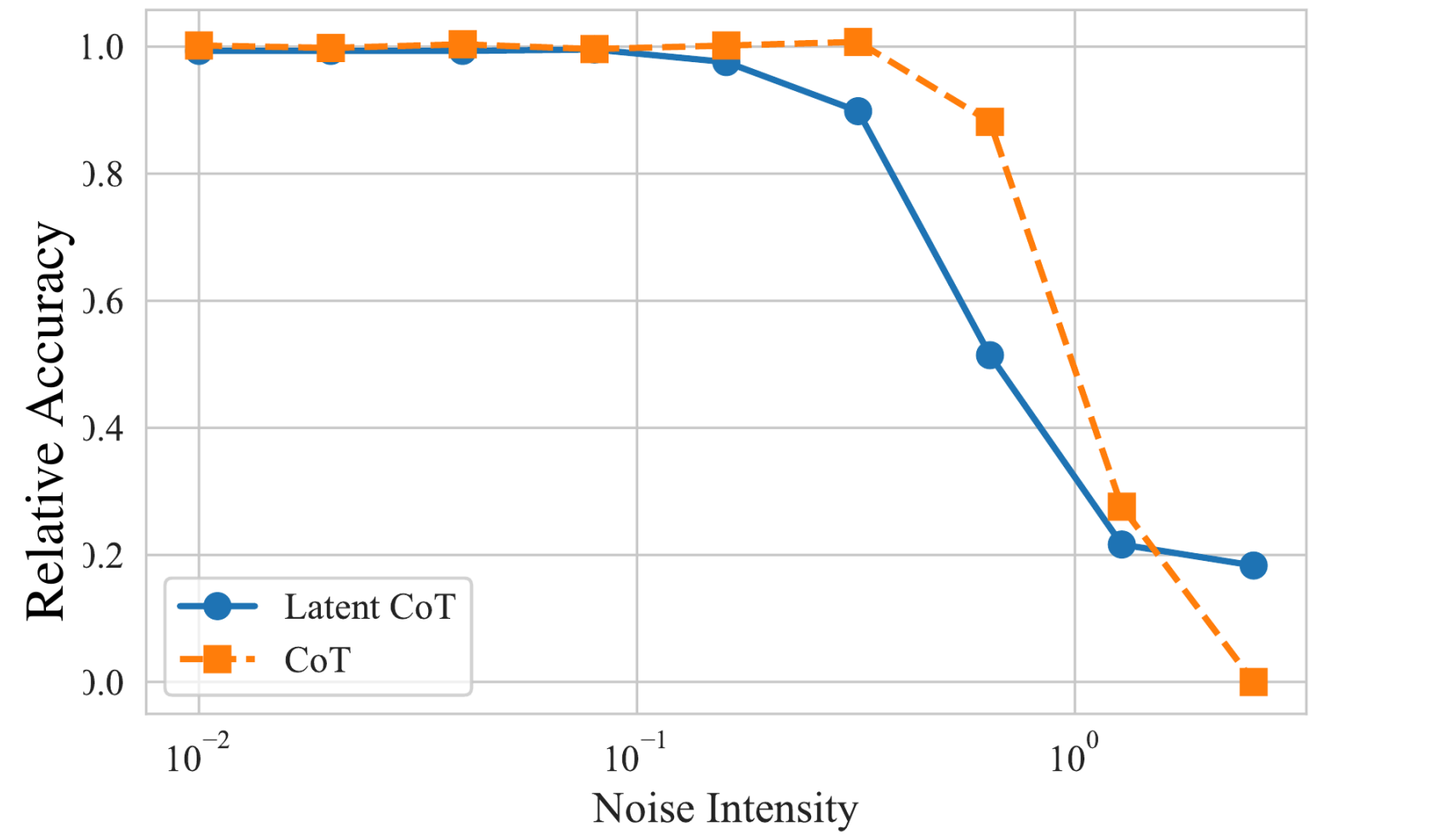


Figure 3 · CoT vs. Coconut on GSM8K.
Observation. CoT maintains a stable plateau at low noise levels; Coconut accuracy deteriorates rapidly as noise increases.
Mechanism analysis. CoT discretization resets sub-decisional perturbations; Coconut's continuous hidden states propagate accumulated error into subsequent steps.

The noise-robustness curves corroborate the stabilizing reset effect of discretization, supporting the theory of continuous-state error accumulation in multi-step computation.

Explore with alternatives. Execute with certainty. Train with expert-aligned scaffolding.

B · Candidate-probability trajectory · ProsQA

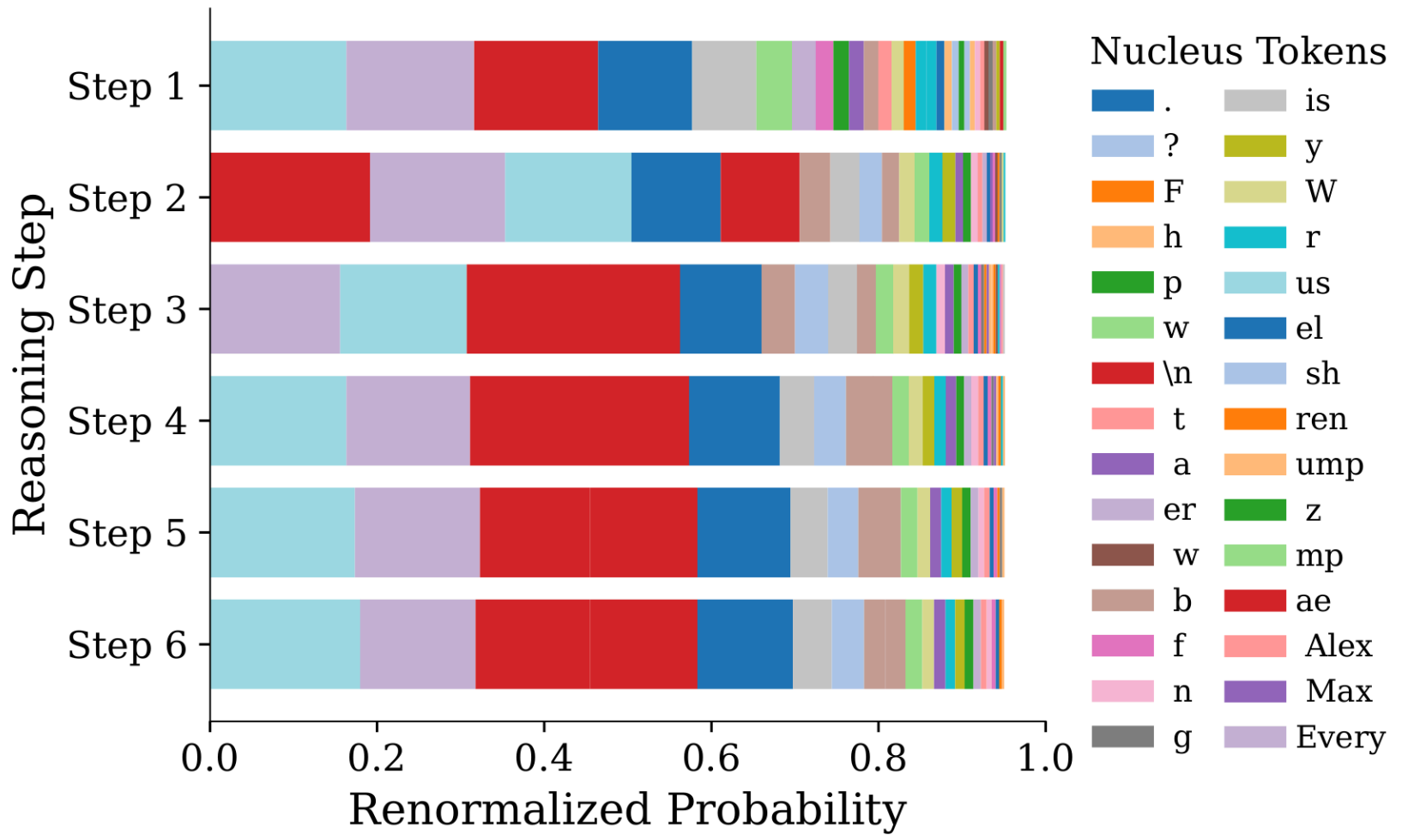


Figure 2 · Six latent reasoning steps on ProsQA.
Every latent step assigns mass to multiple candidates, consistent with dispersed consideration.

D · Task structure determines noise tolerance

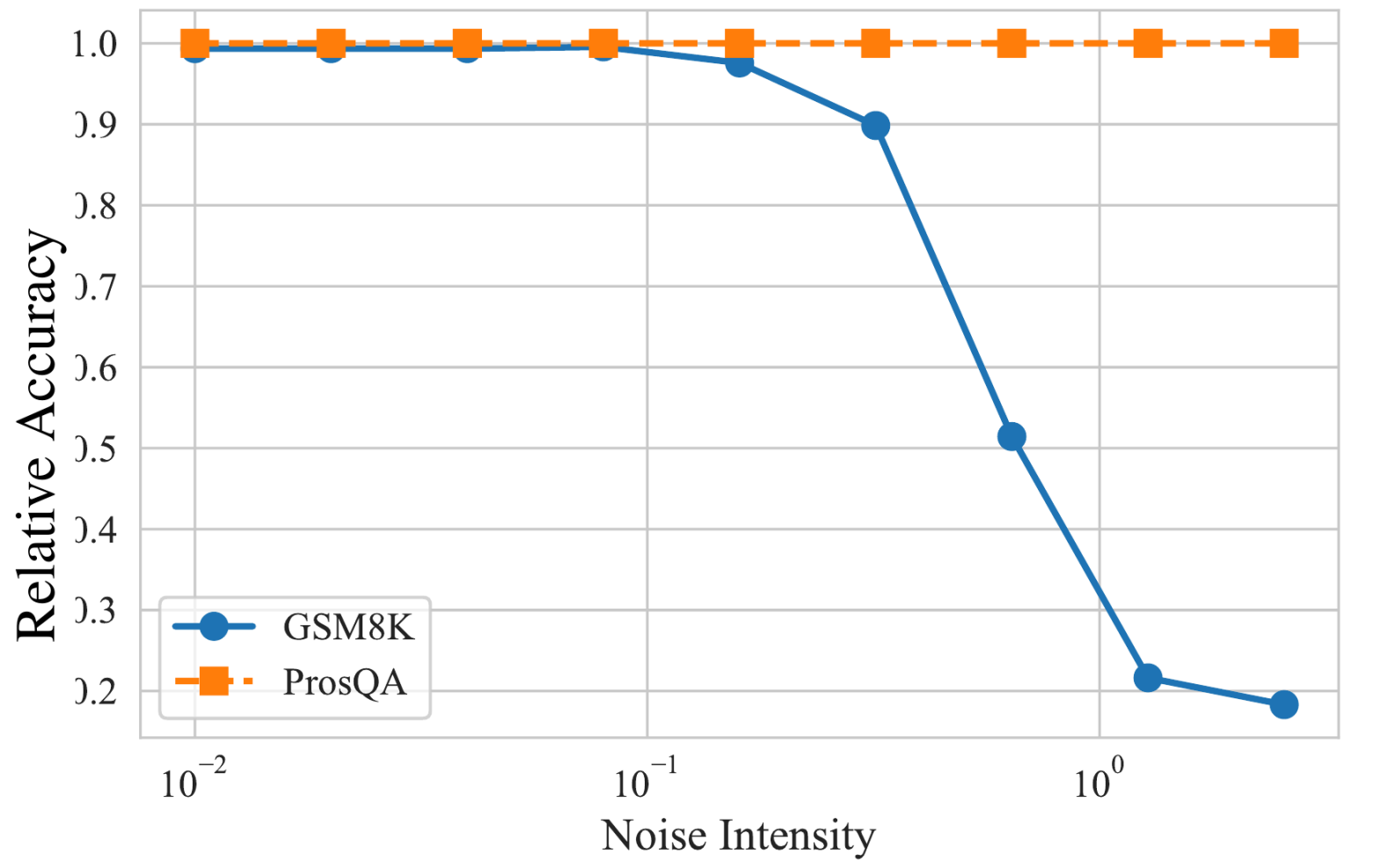


Figure 4 · Coconut on ProsQA vs. GSM8K.
Observation. At the same noise intensity, ProsQA relative accuracy remains nearly stable, while GSM8K accuracy drops substantially.
Impact analysis. The multiple valid solution paths in graph search tolerate larger state deviations; precise computation requires correctness at every step and is highly sensitive to small perturbations.

The effect of continuous-state perturbations depends directly on the task: graph-search reasoning has greater tolerance, while multi-step precise computation is highly vulnerable to accumulated error.

07 Symbolic Index · decisional certainty

$$I_S = \max_{u \in V} p(u \mid h, x)$$

Definition 4.9: $I_S \in [1 / |V|, 1]$. High Index means decisional certainty. CoT uses actual token probabilities; Coconut uses an LM-head probe.

Theorem 4.11 · Symbolic Stability Theorem

The Logit Decision Margin Δ_I is lower-bounded by the Symbolic Index $\mathcal{I}_S$ as follows:

$$\Delta_I \geq \log \left(\frac{\mathcal{I}_S}{1 - \mathcal{I}_S} \right).$$

$\Delta_I = l_{(1)} - l_{(2)}$; $2\|\epsilon\|_\infty < \Delta_I$ preserves argmax. Same softmax support/scale; low Index alone does not prove a small actual margin.

What does a high Index guarantee?	Margin lower bound
Illustrative Index	
0.99	log 99 ≈ 4.60
0.60	log 1.5 ≈ 0.41
Slides 12: illustrative values, not model measurements.	

08 Exploration–Execution Trade-off

More certainty protects the chosen decision, but reduces the mass available to other valid branches.

Higher certainty	Broader exploration
larger guaranteed decision margin	more mass on alternative valid branches

Theorem 4.12 · Exploration-Execution Trade-off Theorem

For a decision with B valid options, the KL divergence from the ideal uniform exploration prior q_{PR} is lower-bounded by:

$$D_{\text{KL}}(q_{\text{PR}} \parallel p) \geq -\log B - \frac{1}{B} \log(I_S) - \frac{B-1}{B} \log \left(\frac{1-I_S}{B-1} \right).$$

This bound is minimized when $I_S = 1 / B$ and grows without bound as $I_S \rightarrow 1$.
Normalize on the same $B \geq 2$ valid branches; this differs from a full-vocabulary probe.

Two operating regimes · slides 21

Explicit CoT: high decisional certainty supports precise execution.

Latent CoT: dispersed candidate mass supports exploration under the stated probability conditions.

E · Measured decisional concentration

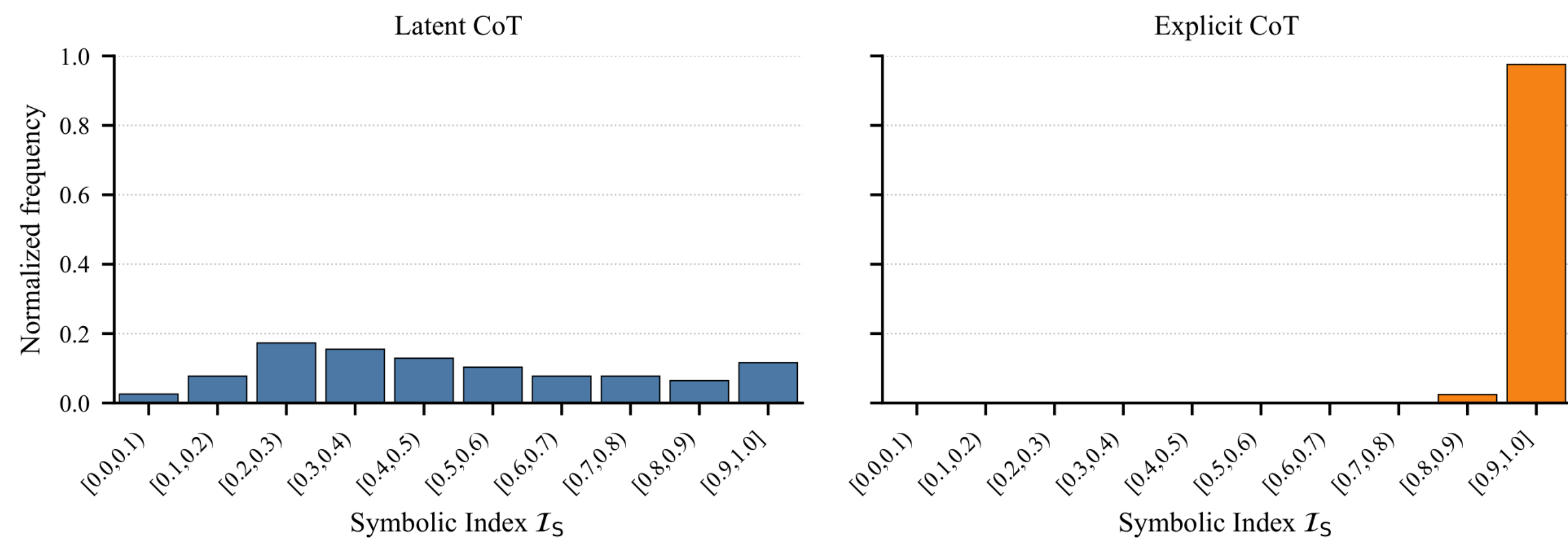


Figure 5 · GSM8K Symbolic Index distribution.
Explicit CoT concentrates in [0.9, 1.0]; Coconut occupies low-to-mid bins. This supports the concentration premise of the trade-off analysis.

Summary & discussion

- Exploration:** CIB and distribution modeling explain the probability-coverage advantage of continuous latent states over multiple valid branches.
- Execution:** Discrete resets and continuous error-accumulation bounds explain the fragility of precise computation.
- EE Trade-off:** The Symbolic Index measures decisional certainty and quantifies the trade-off between exploration and stability.
- Curriculum:** State-distribution bias imposes a lower bound on direct training, motivating curriculum alignment with valid trajectories.

Continuous latent states favor multi-branch probabilistic exploration but lack discrete resets. Curriculum learning is key to overcoming state-distribution bias.

Evidence for latent models is limited to GPT-2-scale Coconut. Causal interventions on the Symbolic Index have not been independently verified.

Setup · slides B4. GPT-2 124M · d = 768 · M = 6 · seed 0. GSM8K test n = 330; ProsQA train/val/test = 2,000/200/300.
C/D · slides 19–20. Gaussian noise: CoT before the LM head; Coconut at every latent step. Metric: Exact-Match Acc(σ)/Acc(0). Curves do not directly measure margins or stepwise error, or locate failure steps.
A/B · slides 18. One sample per task; nucleus p = 0.95, renormalized probabilities. Candidate dispersion does not directly establish valid-path coverage.